One modulus for all binomial digits

Direct quartic equations for syntactic consistency of ZFC
A 332-variable equation and a sparse 333-variable companion

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Abstract

We construct two fixed integer-coefficient quartics whose lack of nonnegative integer zeros is provably equivalent in ZFC to syntactic consistency of ZFC. The first has 332 variables, 263 squared quadratic residuals, 11,823 collected monomials, and coefficient height 85,243,454. The second has 333 variables, 264 residuals, 4,357 monomials, and height 209,508,384. The newest inspected direct-construction baseline has 349 variables, 268 residuals, 12,768 monomials, and height 86,915,503.

The main change replaces complete binomial-numerator powers by Pell residues in a common modulus. A new endpoint argument shows, before interpreting a factorial, that an already required quotient-column modulus dominates every digit modulus. Thirteen of nineteen exact power requests disappear. A second construction arranges the free certificate coordinates to have even indices; it removes the small-ordinate branch without squaring the requested power outputs. This gives a much sparser quartic with one additional variable.

The archive contains every quadratic residual, every expanded coefficient, complete printable equations, portable generators, independent finite checkers, and explicit ordinary proofs. The logical source is the existing calibrated numerical calculus C222. No K215 execution, native coefficient tuple, or Jones universal polynomial is used. The work improves a specific small-coefficient quartic frontier. It does not establish a global simplest equation, a minimum number of variables, or a new end-to-end proof-assistant verification.

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1 The result and what "simplest" means here

The comparison snapshot is ProveIt commit `45b399ec51dfbecb891dd51d8a9e4b6b7c361ac6`. We inspected both the active research and all eight archives then in `docs/incoming`. This matters: the newest incoming direct equation already improves the active review's 399-natural-variable point to 349 variables. The baseline used below is that newer equation, not an earlier published count [1, 2, 3].

Write $\mathbb{N} = \{0, 1, 2, \ldots\}$. An equation in positive integers and an equation in unrestricted integers are different optimization problems; all counts in the following table use *nonnegative integers*. Coefficient height means the maximum absolute coefficient after collecting the full sum of squares, not the largest coefficient in one residual.

Equation	Variables	Rows	Degree	Terms	Height
Newest direct parent	349	268	4	12,768	86,915,503
P_{332}, this work	332	263	4	11,823	85,243,454
P_{333}, this work	333	264	4	4,357	209,508,384

The 332-variable form saves 17 variables, five residuals, 945 monomials, and 1,672,049 units of coefficient height. These are improvements of approximately 4.9%, 1.9%, 7.4%, and 1.9%, respectively, with degree unchanged. The sparse companion saves 16 variables and 8,411 monomials relative to the parent: about 65.9% of its expanded support disappears. Its larger coefficient height is an explicit tradeoff. Compared with P_{332}, it uses one extra variable and 7,466 fewer monomials.

These equations do not supersede every other branch. The repository has nine-positive-variable forms, much smaller variable counts at higher degree, and a 37-integer quartic whose fixed coefficient tuple is very large. This work targets a different measure: an ordinary quartic with small literal coefficients, obtained directly from proof certificates and with no large machine-specific numeral. There is no established total ordering of these measures, and no global optimality theorem is claimed.

Theorem 1.1 (The two delivered consistency equations)**.** *Let $P_{332} \in \mathbb{Z}[z_0, \ldots, z_{331}]$ and $P_{333} \in \mathbb{Z}[z_0, \ldots, z_{332}]$ be the literal polynomials specified in Section 2 and in the two accompanying complete equation documents. Composing the ordinary arithmetic proofs in this report with the pinned C222/refutation calibration gives*

$$\mathrm{ZFC} \vdash \left[\neg\exists \boldsymbol{z} \in \mathbb{N}^{332}\ P_{332}(\boldsymbol{z}) = 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$

and the same statement for P_{333} with 333 variables. No assumption that ZFC is consistent is used in constructing the polynomials or proving either equivalence.

Here Con(ZFC) means absence of a finite syntactic derivation of falsity from the effective ZFC axiom predicate used by ProveIt. The theorem is an ordinary mathematical derivability claim. The delivered finite checks verify the exact polynomials and algebraic transformations. They do not constitute a new Lean, Rocq, or encoded object-language ZFC proof of the entire theorem. The inherited calibration is a substantive, explicit dependency, not a coefficient checksum.

2 A fixed polynomial, rather than an unspecified representation

The recommended equation is exactly

$$\boxed{P_{332}(z_0, \ldots, z_{331}) = \sum_{j=0}^{262} Q_j(z_0, \ldots, z_{331})^2 = 0, \qquad \boldsymbol{z} \in \mathbb{N}^{332}.} \tag{1}$$

Every Q_j is printed in `equation-332.pdf`, generated from `equations/natural332/natural/equation.json`. The full collected polynomial, containing all 11,823 integer coefficients, is `expanded_polynomial.json` in that same directory. The sparse companion is

$$P_{333}(z_0, \ldots, z_{332}) = \sum_{j=0}^{263} \widetilde{Q}_j(\boldsymbol{z})^2;$$

its entire residual list is printed in `equation-333.pdf` and stored under `equations/natural333/`. In the complete equation documents, the residuals of each polynomial are denoted simply by R_j, with the same order and coefficients.

For example, the first three residuals of P_{332}, in the delivered coordinate order, are

$$\begin{aligned} Q_0 &= z_3 - z_0^2 - 24z_0 - 144, \\ Q_1 &= z_4 - z_2 z_3, \\ Q_2 &= z_5 + z_7 - z_2 - z_2 z_6 - z_2 z_7. \end{aligned}$$

Thus the horizon begins as $t = z_0 + 12$, its square is stored in z_3, and the early rows include the factorial ceiling identity. The equation is not just these illustrative rows: all 263 residuals are part of definition (1).

A sparse term `[c,[i,j,...]]` in the JSON means $cz_iz_j\cdots$; repeated indices mean repeated factors and an empty index list denotes a constant. All indices are zero-based, and the coordinate order is fixed. There are no unspecified constants, variable exponents, factorials, binomial coefficients, floor functions, or unbounded products left in either final polynomial. Those functions appear below only to explain the meanings of finite polynomial gadgets.

Every residual has degree at most two. Squaring therefore gives degree at most four. The independent full expansion finds nonzero quartic coefficients, proving that the degree is exactly four on the actual literal coefficients. No random evaluation is used to infer the degree.

2.1 Other domains

The same 332-variable equation over strictly positive integers is $P_{332}(y_0 - 1, \ldots, y_{331} - 1) = 0$. This is a bijective change of domain, although its expanded coefficient height has not been optimized. For an explicit unrestricted-integer equation, put

$$\theta(a, b, c) = a^2 + b^2 + c(c + 1).$$

This polynomial is always nonnegative on integers and represents every nonnegative integer. Indeed the three-square theorem applied to $4n + 1$ gives one odd and two even squares; rearranging gives $n = \theta(a, b, c)$. Consequently

$$P_{332}(\theta(a_0, b_0, c_0), \ldots, \theta(a_{331}, b_{331}, c_{331})) = 0$$

is a completely specified unrestricted-integer companion in 996 variables, of degree eight. It is a domain conversion, not a claim to improve the repository's dedicated integer constructions. The new quartic counts always refer to $\mathbb{N}$.

3 Logical source: a finite certificate of a refutation

The construction uses the calibrated numerical calculus C222 already present in ProveIt. Its formula constructors begin with

$$I(a,b) = 2(a+b)(a+b+1) + 4b.$$

The relation, universal-quantifier, and equality constructors add 1, 2, and 3, respectively. The code 0 is truth and the code 1 is falsity. There are nine fixed axiom/schema constructor polynomials, together with truth, modus ponens, and generalization. The bundle schema has the side conditions $c \neq a, b$, represented by

$$((c-a)(c-b))^2 = (d+1)^2, \qquad d \in \mathbb{N}.$$

This equation is equivalent to both inequalities: when the product is nonzero, take $d = |(c-a)(c-b)| - 1$.

The exact constructor definitions and rule declarations are retained in the vendored `Code254.lean` and `KCode222.lean` files and in the literal source JSON. The relevant calibration interfaces are

$$\mathsf{C222}(1) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZF}}(\bot),$$
$$\mathrm{Ref}_{\mathrm{ZF}}(\bot) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}(\bot).$$

Their exact names are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The second uses the constructible-choice equiconsistency transport. The supplied source pin and hashes identify these existing statements [6]. They do not involve a machine execution hypothesis.

This use of a direct proof calculus is consistent with the broader program of arithmetic representations of provability studied by Carl and Moroz and, more recently, Formisano, Gavazzi, and Omodeo [7, 8]. The specific source and optimizations here are the ones identified in the ProveIt provenance; those papers are context, not substitutes for this source's calibration.

3.1 One factorial serves two purposes

Choose a horizon $t \geq 12$, eventually the number of padded proof steps, and let $F = t!$. Every formula label in a local step is required to be at most t. Define its weight by

$$w_j = 1 + jF, \qquad 0 \leq j \leq t.$$

For distinct positive $j, k \leq t$, these weights are coprime. A common prime divisor would divide $(j-k)F$, would not divide F, and would therefore divide $j-k$; but that prime is at most t, hence divides $t!$, a contradiction. Truth has the harmless weight $w_0 = 1$.

A local rule with premise labels p, q and conclusion v is encoded by

$$S' w_v = S w_p w_q. \tag{2}$$

Unused premises are truth. The accumulator starts at $S_0 = w_1$ and ends at $S_t = 1$. Since every weight is positive, its local equalities propagate positivity of all accumulator entries. Multiplication of (2) and cancellation give

$$\prod_{i<t} w_{v_i} = w_1 \prod_{i<t} w_{p_i} w_{q_i}.$$

Prime valuations isolate each positive label and give

$$\#\{i : v_i = j\} - \#\{i : p_i = j\} - \#\{i : q_i = j\} = [j = 1]. \tag{3}$$

These balances force a derivation even if the certificate's rule multiset contains cycles. Let D be the closure of truth under its valid rule instances. If falsity were outside D, every rule with conclusion outside D would have a premise outside D. Summing (3) over the finitely many positive labels outside D would give a nonpositive left side and the right side 1. Hence falsity is in the closure. Conversely, expanding a finite proof tree backwards and storing the product of the weights of its pending obligations gives the required positive accumulator. Truth steps pad it to any sufficiently large horizon.

The same factorial also gives pairwise-coprime reciprocal moduli

$$d_i = 1 + \frac{F^2}{i} \quad (1 \le i \le t), \qquad B = \prod_{i=1}^{t} d_i = \binom{F^2 + t}{t}.$$

Each $d_i \equiv 1 \pmod F$, so its prime divisors exceed t. A common prime divisor of d_i, d_j would divide $i - j$, again impossible. The divisions and product in this paragraph describe the proof; the actual graph uses the displayed binomial identity.

3.2 Amplified CRT and a derived label range

Set

$$R = 2 + t + x + y + F, \qquad K = 2R, \qquad N = B^K.$$

A beta code (x, y) stores the overlapping accumulator entries with moduli $1 + (i + 1)y$. Eight CRT columns store the four local rule parameters, two accumulator coordinates, and two quotient coordinates. A ninth representative stores the row index. Their range conditions have the form $N \mid \binom{a}{k}$.

The key range implication is

$$p^e \mid \binom{a}{k} \quad \Longrightarrow \quad a \bmod p^e < k. \tag{4}$$

For $a \ge k$, if $[a - k + 1, a]$ contains no multiple of p^e, then the valuation formula for the binomial has zero terms from level e onwards and at most $e - 1$ remaining units. That contradicts the assumed divisibility. The case $a < k$ is immediate. Only this direction is used; its converse is generally false.

To recover a row, choose a prime of its reciprocal modulus, use (4) to decode all bounded coordinates, and reduce the local nonnegative source polynomial modulo the corresponding amplified prime power. Its absolute size is below that prime power, so its zero residue is an actual zero. Conversely, CRT at precision d_i^{2K} gives representatives whose falling factorials have enough valuation even after division by $k!$. The factor 2 in the completeness precision is essential.

Instead of another local label-bound coordinate, define J as the positive polynomial sum of all nine schema codes and all labels that could occur in the chosen local rule. Require $N \mid \binom{J}{t+1}$. A separate majorant bounds this derived expression. It must not be included in the upper bound on independently chosen CRT representatives. This is the established derived-range architecture of the newest parent [2]; both delivered equations retain that distinction.

4 The new dominant denominator

The old arithmetic graph evaluates complete powers $(q + 1)^n$ to extract a small binomial digit. We retain only what the digit actually uses: the residue of that power in a suitable modulus. Package 108 already introduced the cofinal Pell method for such residues [4]. The new issue is to combine it with the latest factorial architecture without paying for a new large denominator power.

The semantic endpoint equations are

$$\begin{aligned} x &= F + 1 + (y+1)u_0, \\ y &= F + r_0 + 1, \\ x &= 1 + [1 + (t+1)y]u_t. \end{aligned} \tag{5}$$

Before its factorial meaning is known, the proposed output F occurs in the ceiling certificate

$$\begin{aligned} \rho &= Ft^2, & v &= (\rho+1)^t, \\ v + c &= cF + r + 1, & c &= r + s + 1. \end{aligned} \tag{6}$$

Initially c is only a proposed value of $\binom{\rho+t}{t}$. All witnesses here are nonnegative.

Lemma 4.1 (Endpoint dominance before factorial interpretation)**.** *Once the retained exact power in (6) has been recovered, the raw polynomial rows imply*

$$F \geq 1, \qquad y \geq 2, \qquad x \geq 2 + (t+1)y > \max(t+1, y+1).$$

No assertion that $F = t!$ or that c is a binomial is needed.

Proof. The power gives $v > 0$. If $F = 0$, the last two equations of (6) give $v = -s \leq 0$. Thus $F \geq 1$, and (5) gives $y \geq 2$ and $x \geq 2$. Its final equation forces $u_t \geq 1$, giving the claimed lower bound for x. The strict comparisons use only $t \geq 1$ and $y \geq 2$. □

The radix is

$$q = N(N + v + 1)^K. \tag{7}$$

Every lower digit index of the smaller construction belongs to $\{t, t+1, x+1, y+1\}$. Exact binomials use lower index t. Retain exact powers q^t, q^{x+1}, q^{y+1} and define $q^{t+1} = q\,q^t$. The single common modulus is

$$\boxed{M = Nq^{x+1}.} \tag{8}$$

This is already the modulus required for the two quotient-column range tests. By Lemma 4.1, every range modulus Nq^k divides M; the exact-digit modulus q^{t+1} does too. The proof uses no unvalidated factorial to establish the comparison.

The six remaining exact powers are

$$(Ft^2 + 1)^t, \quad B^K, \quad (N + v + 1)^K, \quad q^t, \quad q^{x+1}, \quad q^{y+1}.$$

All twelve binomial numerator powers disappear. Replacing q^{t+1} by a multiplication removes one further power request, giving nineteen down to six. The numerator indices still have certified Pell pairs; only their full power outputs disappear.

5 Pell residues with one cofinal choice of parameter

For $A \geq 2$, write

$$X_n(A) + Y_n(A)s = (A + s)^n, \qquad s^2 = A^2 - 1.$$

The integer sequences start at $(1, 0)$, then $(A, 1)$, and follow the usual multiplication recurrence. The classification and index properties used here are the standard Pell ingredients of the Diophantine representation of powers [9]; the exact shared index certificate is retained in the vendored source.

5.1 A lower bound without fixing the parameter

Let W be 2 plus the sum of distinct semantic atoms appearing as bases, exponents, or proposed outputs of the six exact-power requests. Impose

$$Z = z + 1, \quad L^2 - \big((W+1)^2 - 1\big)(WZ)^2 = 1, \quad A = L + \sigma, \qquad z, \sigma \in \mathbb{N}. \tag{9}$$

Pell classification gives $L = X_e(W+1)$ and $WZ = Y_e(W+1)$. The recurrence modulo W gives $Y_e(W+1) \equiv e \pmod{W}$. Thus $e \geq W$, and

$$A \geq L \geq (W+1)^W > W^W. \tag{10}$$

Crucially, after selecting this lower bound, *any* larger A is available through σ.

For a proposed exact power $m = n^k$, the graph imposes

$$T_n = 2An - n^2 - 1, \qquad X_k + nY_k = AY_k + m + T_n h.$$

Its nonnegative modulus row excludes $n = 0$. The frame bound makes $T_n > m, n^k$, while the Pell recurrence gives $X_k + (n - A)Y_k \equiv n^k \pmod{T_n}$. The residue is therefore exact: $m = n^k$. This recovers the retained powers before interpreting binomials or factorials.

5.2 One root congruence for every numerator

Put $b = q + 1$. Impose

$$2Ab = b^2 + 1 + Mh_0, \qquad h_0 \in \mathbb{N}. \tag{11}$$

Because $N \mid q$, we have $\gcd(b, M) = 1$. The linear congruence for A is soluble: $\gcd(2b, M)$ is either 1 or 2, and in the second case q is even, so $b^2 + 1$ is even. Its solutions form an arithmetic progression. We may choose A in that progression above both L and b, which also makes h_0 nonnegative. This is compatible with all finitely many shared index witnesses. All genuine semantic data are chosen before this A.

Equation (11) makes the substitution $s \mapsto b - A$ a homomorphism to $\mathbb{Z}/M\mathbb{Z}$, because

$$(b - A)^2 - (A^2 - 1) = b^2 - 2Ab + 1 \equiv 0 \pmod{M}.$$

It follows, for every $n \geq 0$, that

$$\boxed{X_n + (b - A)Y_n \equiv b^n \pmod{M}.} \tag{12}$$

This identity is the reason an enormous numerator need not be produced.

For $n \geq 1$, the norm also gives

$$X_n^2 - (A-1)^2 Y_n^2 = 1 + 2(A-1)Y_n^2 > 0.$$

Hence

$$X_n + (b - A)Y_n > (b-1)Y_n > 0. \tag{13}$$

The positivity is needed for completeness over $\mathbb{N}$; merely proving a congruence would not justify natural quotient witnesses.

5.3 Extracting the actual digits

Let $d = q^k$. An exact binomial digit c is represented by

$$X_n + bY_n = AY_n + r + dc + qdh,$$
$$d = r + s + 1, \qquad q = c + o + 1. \tag{14}$$

Thus $0 \le r < d$ and $0 \le c < q$. Since $qd \mid M$, (12) identifies $r + dc$ as the least residue of $(q+1)^n$ modulo q^{k+1}. When the low coefficients $\binom{n}{j}$, $0 \le j \le k$, are smaller than q, the binomial expansion has no carries there. Unique quotient and remainder then give $c = \binom{n}{k}$.

For a range test only divisibility by N is needed. Replace (14) by

$$X_n + bY_n = AY_n + r + Ndh, \qquad d = r + s + 1. \tag{15}$$

Because $N \mid q$, the residue of $(q+1)^n$ modulo Nd has the form

$$r_0 + d\left(\binom{n}{k} \bmod N\right), \qquad 0 \le r_0 < d.$$

It lies below d exactly when $N \mid \binom{n}{k}$. This proves the desired range semantics. Conversely choose the true digits and remainders. The positive integer in (13) has the same least residue, so its quotient is nonnegative. Thus every displayed new row has witnesses in the required domain.

6 Bootstrap order and the finite horizon constant

The dependencies cannot be rearranged arbitrarily. In particular a factorial estimate must not be used to validate the digit that first establishes that factorial. The soundness order is:

1. Restore the raw gate system from its checked elimination certificates.
2. Recover the exact powers through the cofinal frame and shared Pell indices. Zero power bases are excluded by their raw modulus rows.
3. Apply Lemma 4.1; establish that every requested digit modulus divides M, without factorial semantics.
4. Apply the common root congruence and the Pell residue identity.
5. Validate the first exact binomial digit, recover $F = t!$, then validate the reciprocal product binomial and all range digits.
6. Decode bounded rows by amplified CRT, extract a C222 derivation by the finite Horn balance argument, and use the pinned calibration.

For the first digit, exact powers give $q > v^2$ and $v = (Ft^2+1)^t$. After $F \ge 1$ is known from raw rows, the low coefficients of $(q+1)^{Ft^2+t}$ are bounded by $v < q$. The ceiling identity (6) then yields $F = t!$. Next $F^2 + t \le (Ft^2+1)^2$, so the low coefficients of the reciprocal product digit are at most $v^2 < q$. This establishes $B \ge 2$ and $N \ge 2^K$ before the remaining range estimates.

The parent majorants use

$$E = 8,388,608, \qquad \ell = 4,194,304.$$

They bound the local source by R^E, and the derived label polynomial on bounded inputs by $(n+1)^\ell$. These are proof bounds, not additional coefficients or unknowns in the final equation. The older horizon floor 16 can be lowered to 12 because

$$\begin{aligned} 12! &= 479,001,600 > 2^{28}, \\ 28E &= 234,881,024 < 2^{28}, \qquad 2E < 2^{28}, \\ 2 \cdot 12! &= 958,003,200 > 3\ell \cdot 13 = 163,577,856. \end{aligned}$$

At $r = 2^{28}$ these inequalities give $r^E < 2^r$. For larger integers r,

$$(1 + 1/r)^E \le \sum_{j=0}^{E} (E/r)^j < 2,$$

so induction preserves the bound. Hence, after factorial interpretation, $R^E < 2^R < 2^K$. The last displayed factorial inequality similarly propagates to $2t! \ge 3\ell(t+1)$, supplying the derived-digit bound. Completeness can always pad a finite proof beyond this minimum; no finite proof length is being bounded in advance.

7 A sparse companion by arranging even indices

The second contribution changes the certificate representation itself. An earlier incoming method doubled every power exponent and replaced $m = n^k$ by $m^2 = n^{2k}$ to remove exceptional Pell-index cases [5]. Here we arrange the free source coordinates so that the *actual* exponents are already even. The power outputs remain unsquared. This is an existence-preserving change of certificates, not a pointwise substitution into every old witness.

The free outer coordinates become

$$t = 2t_0 + 12, \quad x = 2x_0 + 2, \quad F = 2f_0, \quad y = F + 2r_0 + 2. \tag{16}$$

Every independently chosen CRT representative is $2(a_0 + 1)$. Its aggregate bound doubles from $9N^2$ to $18N^2$. The four local parameter bounds are now strict below t, the two accumulator bounds strict below y, and the two quotient bounds strict below x. Accordingly the lower range indices are t, y, x, without the old additions of 1. The derived label range is

$$N \mid \binom{2J}{2t}. \tag{17}$$

Its decoding gives $2J < 2t$, hence $J < t$. All required upper and lower Pell indices are syntactically even before any factorial or binomial meaning is invoked. In particular $F = 2f_0$ is enforced as a polynomial definition; parity is not inferred prematurely from the intended factorial.

7.1 Why the parity restrictions preserve completeness

Choose an even padded horizon strictly above all finitely many source parameters and label sums, then set $F = t!$. It is even. Choose an even y, divisible by $t!$, strictly larger than all accumulator values and at least $F + 2$. The beta moduli $1 + (i+1)y$ are odd. Their joint CRT solution can therefore be made positive and even by adding a multiple of their odd product; make it large enough to obtain the

prescribed $x = 2x_0 + 2$. Its quotient columns are strictly below x because the relevant positive divisors exceed 1.

The reciprocal moduli d_i are odd as well. A residue class modulo their product at precision $2K$, namely N^2, has a positive even representative at most $2N^2$. Taking all nine representatives this way establishes the aggregate bound. Multiplication by 2 in (17) is compatible with the same high-precision CRT lifting. It is not an appeal to the false converse of (4).

The ordinary ranges satisfy the old carry estimates with 18 in place of 9. For the derived expression, the separate majorant gives $2J \le N^{3\ell+1}$. It suffices that

$$2t(3\ell + 1) \le 2F < K.$$

At $t = 12$, its left half is $12(3\ell + 1) = 150,994,956 < 12!$; multiplication by successive integers preserves the required factorial inequality. This verifies the stronger derived-label digit budget independently of the aggregate bound.

7.2 Even-index exclusion, with no hidden sign assumption

For each seed index k, the shared certificate has main and auxiliary Pell equations and congruences

$$X^2 - (A^2 - 1)Y^2 = 1, \qquad S^2 - (B_*^2 - 1)T^2 = 1,$$
$$S \equiv X \pmod{U}, \qquad T \equiv k \pmod{4Y}.$$

The common parameters satisfy

$$U^2 - (A^2 - 1)V_*^2 = 1, \quad V_* \ge 1, \quad B_* \equiv A \pmod{U}, \quad B_* \equiv 1 \pmod{4V_*}, \quad Y^2 \mid V_*.$$

The literal product defining V_*, together with its strict positive definition, forces every seed $Y \neq 0$.

If $Y = 1$, the main norm gives $X = A$. The standard abscissa residue lemma says

$$X_j(A) \equiv \pm A \pmod{X_h(A)} \quad \Longrightarrow \quad j \text{ is odd.}$$

One proof reduces the index modulo $2h$ using $(A + s)^{2h} \equiv -1 \pmod{X_h(A)}$, reflects into $[0, h]$, and uses strict increase of X_j to force the reduced index to 1. Apply this to the auxiliary Pell pair. Its index is odd, and its odd base B_* makes its ordinate odd. But $T \equiv k \pmod{4Y}$ and the even k make that ordinate even. Contradiction. Thus $Y \ge 2$, and Pell growth gives $Y \ge 2A > k$.

For the last inequality, the sparse construction includes every modular-only upper index in W as well as the exact-power atoms. Omitting those extra indices would leave an unjustified bound for the large derived expression $2J$. With the complete frame, the ordinary index criterion now recovers the requested index without a separate $Y - k \ge 0$ coordinate. All variables are natural throughout this argument. It is not a signed-domain proof and cannot be transferred to unrestricted integers by simply dropping the same guards.

The dominant-modulus argument strengthens under (16): raw rows give $F \ge 2$, $y \ge 4$, and $x \ge 2 + (t + 1)y > 2t, y$. Hence the sparse common modulus is

$$M = Nq^x.$$

Its extra denominator is obtained by $q^{2t} = (q^t)^2$, so it too has only six exact-power requests. The common root, positive residue, and digit proofs then apply unchanged. The complete quantified proof and literal parity checker are included separately in the archive.

8 Exact verification and the proof boundary

The finite verification has two distinct purposes. First, it binds the mathematical formulas to the actual emitted rows. Second, it checks the full coordinate eliminations and every final coefficient. Neither purpose is a bounded search for a refutation or a test of consistency.

For P_{332}, the independent checker restores all 914 original quadratic rows through 496 discarded gate coordinates, checks 155 further monic eliminations, and expands all 11,823 coefficients. The two restorations with negative coefficients have explicit rational sum-of-squares nonnegativity certificates; coefficient signs are not silently treated as a positivity proof. The unchanged 441-row source prefix is compared with the pinned parent, apart from the recorded horizon constant. A second checker verifies all 47 new macro identities over exact sparse polynomial rings. Deliberate changes to a restoration coefficient and a polynomial coefficient are rejected.

For P_{333}, the same independent algebraic checker handles 922 original rows, 516 gate restorations, 142 further monic contractions, and all 4,357 final coefficients. Its separate structure check verifies the syntactic parity, actual altered source bounds, complete frame, shared positive ordinate, dominant modulus, and absence of deleted index-gap rows. The ordinary parity and CRT proof remains necessary beyond these finite checks.

The archive retains the ordinary source proofs, calibration declarations, source identities and licensing notices. The complete literal equations and their checkers can be used without network access. Generator replay uses only the supplied Python modules and the standard library; LaTeX is needed only to rebuild the PDFs. The verification receipts state their actual scope. No test output is relabeled as a new proof-assistant certificate.

8.1 Why nonnegative elimination is checked explicitly

An affine gate such as $z = x + y$ can be deleted over $\mathbb{N}$ because its restoring value is always nonnegative. A general monic row $z = f(\boldsymbol{x})$ cannot be deleted merely because it has coefficient 1 at z: the polynomial f might take negative values. The supplied certificates give a coefficientwise nonnegative formula or a verified nonnegative square decomposition for every restoration used. Each deleted coordinate is uniquely restored, and all retained rows are checked after substitution. This is the finite step that carries the high-level arithmetic construction to the stated exact variable counts.

9 What this round establishes, and what remains open

The completed contribution is a pair of explicit consistency equations and their proof-preserving compiler changes. The new endpoint dominance lemma removes a separate common-denominator power; the combined modular construction removes thirteen exact-power requests; parity padding gives a sparse alternative with unsquared power outputs. The repository-relative comparison includes the latest incoming archives. The generic Pell residue identity, CRT, coprime label weights, and even-index obstruction are credited to their existing sources rather than claimed as newly discovered general theorems.

Several tempting shortcuts were not used. The common Pell parameter is chosen afresh during completeness, so the proof does not falsely retain an old frame that bounded much larger numerator outputs. The derived label expression is not squeezed into the bound on independent CRT representatives. Parity is established by literal constructors before factorial interpretation. Signed modular-only Pell

pairs are not assumed to have the orientation supplied by a removed exact-power call.

The next meaningful reductions would require a further change in the certificate or index architecture, or a sound reduction of its local source polynomial. Removing another positivity condition without a replacement argument is not such a reduction. Likewise, combining two radicals into one square can introduce nontrivial squareclass solutions; finite searches do not justify discarding those branches. Those approaches remain research questions, and no unproved candidate is included among the delivered consistency equations.

This is measurable progress on a small-coefficient quartic representation of ZFC refutability. It supplies two new explicit points for comparison; it does not supply a lower bound establishing global optimality for either one.

A Reading and rebuilding the package

Location	Purpose
`report.tex`, `report.pdf`	This research article.
`equation-332.tex/.pdf`	Every residual of the 332-variable polynomial, in order.
`equation-333.tex/.pdf`	Every residual of the sparse companion.
`equations/natural332/`	Raw system, exact projection, all residuals and collected coefficients.
`equations/natural333/`	Corresponding complete sparse-system artifacts.
`code/`	Portable producers and independent checkers.
`proofs/`	Complete ordinary transformation proofs and independent review.
`vendor/round14/`	Pinned prior architecture, source calculus and arithmetic dependencies.
`verification/`	Actual finite verification and reproduction receipts.
`SOURCE_LEDGER.json`	Compared repository snapshot, archive identities and inherited source map.
`MANIFEST.json`	SHA-256 and byte length of every packaged file except the manifest itself.

The top-level `verify_all.py` verifies the manifest and both equations, including their independent algebraic and structural checks. Use `python3 verify_all.py -rebuild` to generate both equations again in a temporary directory and compare every selected equation and coefficient file. The PDF rebuild command is `latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex`; the two complete equation documents compile the same way. See the package README for exact commands and the interpretation of each verification.

References

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[2] ProveIt incoming research package, *Powered Saturation and Factorial Carriers, Round 14*, `ConZFC_Powered_Saturation_Factorial_Carriers_Round14_2026-10-24.zip`, in `docs/incoming` at [1]. The used components are `architecture/DERIVED_LABEL_RANGE_PROOF.md`, `FACTORIAL_CARRIER_PROOF.md`, and `HYBRID_PROOF.md`.

[3] ProveIt, *Later arrivals and current size reductions*, `PROPOSAL_REVIEW_LATEST.md`, at [1].

[4] ProveIt package 108, *Modular binomial certificates through a cofinal Pell frame*, `108-pell-residues/proofs/modular_binomial.txt`, at [1].

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